

Name : _____

Teacher/ Class : _____

SYDNEY TECHNICAL HIGH SCHOOL

**HSC ASSESSMENT TASK 1**

DECEMBER 2006

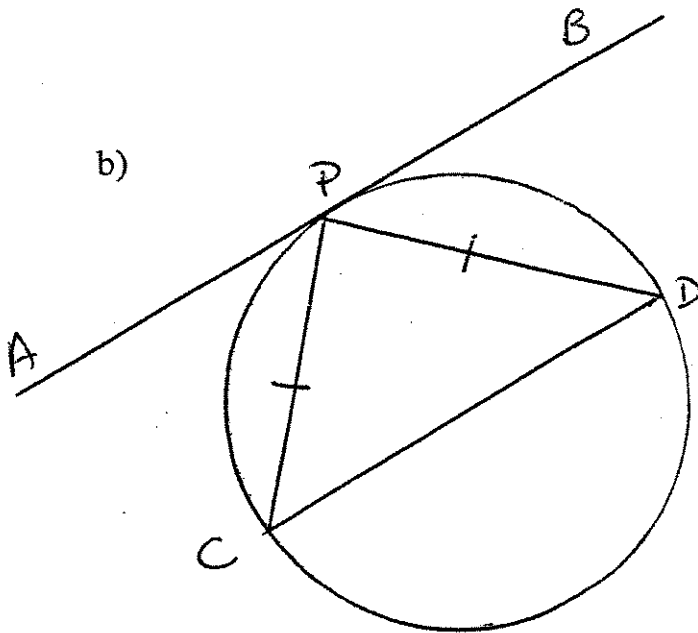
MATHEMATICS - EXTENSION 1Time Allowed : **70 minutes****Instructions:**

- Write your name and class at the top of each page.
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- Start **each** question on a new **page**.
- Diagrams unless otherwise stated are not to scale.

Q1	Q2	Q3	Q4	Q5	Q6	TOTAL
/8	/5	/8	/10	/10	/9	/50

Question 1

- a) The sum of an infinite geometric series is $\frac{3}{2}$. If the common ratio is halved the sum of the resulting infinite series is $\frac{12}{17}$. Find the first term and common ratio of the original series. (4 marks)



PC and PD are equal chords of a circle. A tangent AB is drawn at P . Prove that AB is parallel to CD .

(4 marks)

Question 2 (Start a new page)

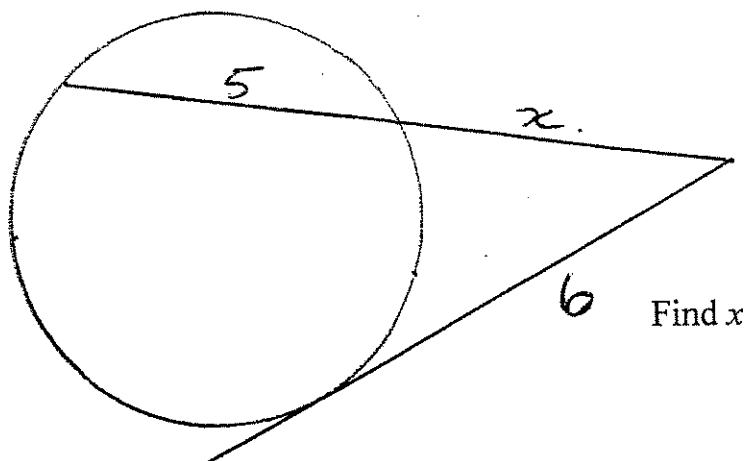
- a) The n th term of a sequence is given by

$$T_n = a\left(\frac{1}{2}\right)^n + bn$$

If the first 3 terms are 11, 10, 11 find a and b , and hence the fourth term.

(3 marks)

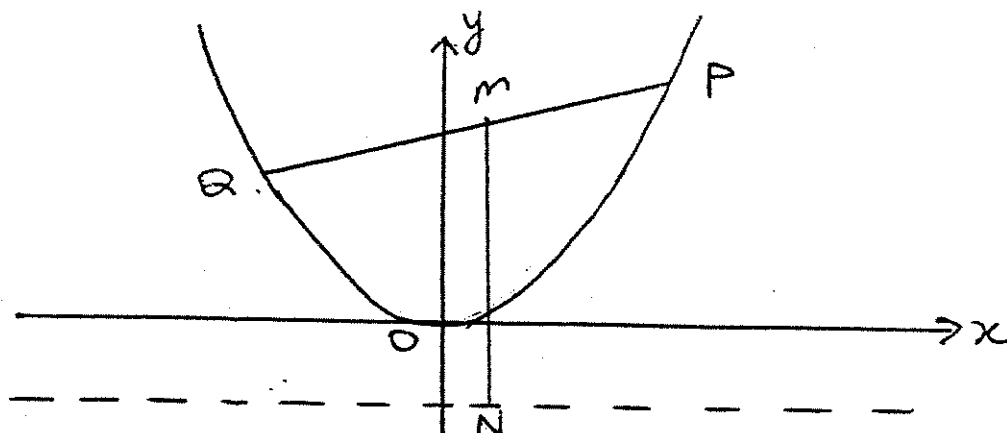
b)



Find x

(2 marks)

Question 3 (Start a new page)



Let $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ be points on the parabola $x^2 = 4ay$ as shown in the diagram.

a) Show that the equation of PQ is

$$y = \frac{p+q}{2}x - apq$$

(2 marks)

b) Show that if the chord PQ passes through the focus $(0, a)$,

then $pq = -1$

(1 mark)

- c) M is the midpoint of the focal chord PQ and N lies on the directrix vertically below M . T is the midpoint of MN .

Write down

i) the co-ordinates of M

(1 mark)

ii) the co-ordinates of N

(1 mark)

iii) show that T has co-ordinates

$$\left[a(p+q), \frac{a}{4}(p^2+q^2-2) \right]$$

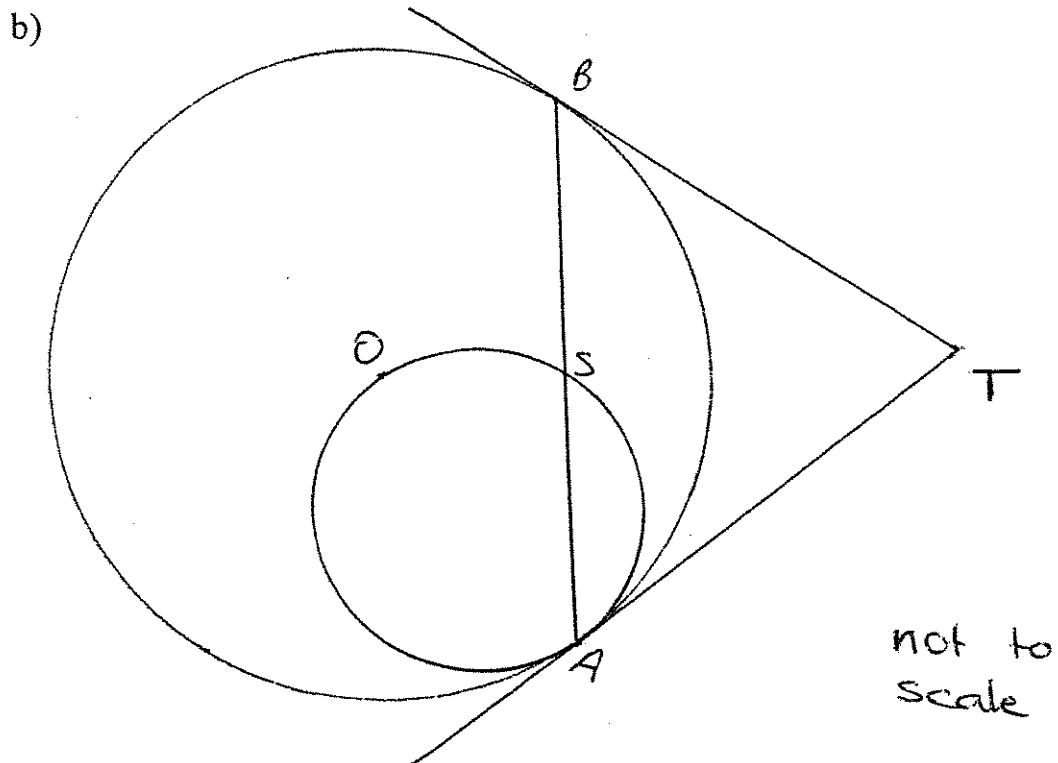
(1 mark)

iv) show that the locus of T is $x^2 = 4ay$

(2 marks)

Question 4 (start a new page)

- a) The sum of three consecutive terms of an arithmetic series is 21, and the sum of their squares is 155. Find the three terms by letting a be the middle term. (5 marks)



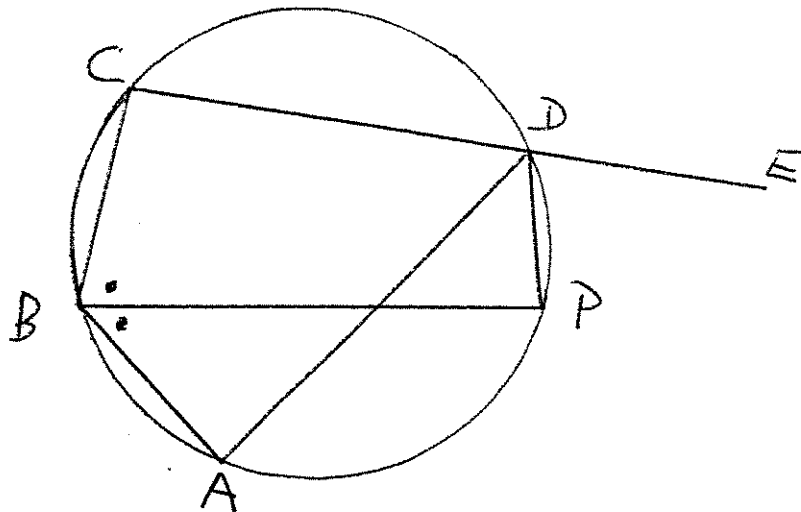
Two circles touch internally at a point A and the smaller of the two circles passes through O , the centre of the larger circle. AB is any chord of the larger circle, cutting the smaller circle at S . The tangents to the larger circle at A and B meet at a point, T .

- Prove i) AB is bisected at S (3 marks)
 ii) O, S and T are collinear (2 marks)

Question 5 (start a new page)

- a) The normal at any point $P(2at, at^2)$ on the parabola $x^2 = 4ay$ cuts the y axis at Q and is produced to a point R such that $PQ = QR$
- show that the equation of the normal is $x + ty - at^3 - 2at = 0$ (1 mark)
 - find the co-ordinates of Q (1 mark)
 - write down the coordinates of R (2 marks)
 - by eliminating t show that the locus of R is $x^2 = 4a(y - 4a)$ (2 marks)

b)



In the diagram $ABCD$ is a cyclic quadrilateral. CD is produced to E .
 P is a point on the circle such that $\angle ABP = \angle PBC$

- copy the diagram
- give a reason why $\angle ABP = \angle ADP$ (1 mark)
- show that PD bisects $\angle ADE$ (2 marks)
- if, in addition, $\angle BAP = 90^\circ$ and $\angle APD = 90^\circ$ state where the centre of the circle is located. (1 mark)

Question 6 (start a new page)

A man borrows \$30 000 at 12 % p a compound interest. If the principal plus interest are to be paid by 20 equal annual instalments,

- i) Write an expression for A_1 the amount owing after 1 year. Let the annual instalment be M . (1 mark)
- ii) Show that the amount owing at the end of 2 years is given by
$$A_2 = 30\,000 (1.12)^2 - M(1.12 + 1)$$
(1 mark)
- iii) Find the annual instalment (3 marks)

b) Prove by mathematical induction that

$$\sum_{r=1}^n r^2 = \frac{1}{6}n(n+1)(2n+1) \quad (4 \text{ marks})$$

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Mathematics - Extension I

December 2006.

Question 1

$$a) \quad \frac{a}{1-r} = \frac{3}{2} \quad \Rightarrow \quad 2a = 3 - 3r \quad (1)$$

$$\frac{a}{1-\frac{1}{2}} = \frac{12}{17} \quad \Rightarrow \quad 17a = 12 - 6r \quad (2)$$

$$(1) \times 2 \quad 4a = 6 - 6r \quad (3)$$

$$(2) - (3) \quad 13a = 6$$

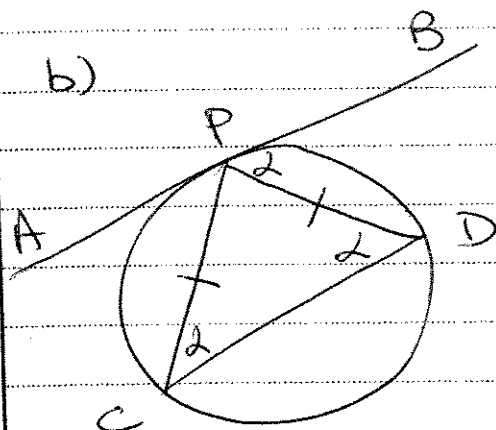
$$\therefore a = \frac{6}{13}$$

Put $a = \frac{6}{13}$ into (1)

$$\frac{12}{13} = 3 - 3r$$

$$3r = \frac{27}{13}$$

$$r = \frac{9}{13}$$



$\angle BPD = \angle PCD$ (angle in the alternate segment)

$\angle PDC = \angle PCD$ (base angles of an isosceles triangle)

$$\therefore \angle BPD = \angle PDC$$

Since a pair of alternate angles are equal $AB \parallel CD$.

Question 2

$$a) \quad T_n = a\left(\frac{1}{2}\right)^n + bn$$

$$T_1 : a\left(\frac{1}{2}\right) + b = 11$$

$$\text{ie } a + 2b = 22 \quad (1)$$

$$T_2 : a\left(\frac{1}{2}\right)^2 + 2b = 10$$

$$\text{ie } a + 8b = 40 \quad (2)$$

$$(1) - (2) \quad -6b = -18$$

$$b = 3$$

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Put $b=3$ into ①

$$a + b = 22$$

$$\therefore a = 16$$

$$\begin{aligned} T_4 &= 16\left(\frac{1}{2}\right)^4 + 3(4) \\ &= 1 + 12 \\ &= 13 \end{aligned}$$

$$b) \quad x(5+x) = 6^2$$

$$5x + x^2 = 36$$

$$x^2 + 5x - 36 = 0$$

$$(x+9)(x-4) = 0 \Rightarrow x = -9 \text{ or } x = 4$$

But x must be positive

$$\therefore x = 4$$

Question 3

a) Using the two point form

$$\frac{y - ap^2}{x - 2ap} = \frac{ap^2 - aq^2}{2ap - 2aq}$$

$$= \frac{a(p-q)(p+q)}{2a(p-q)}$$

$$\begin{aligned} p &\neq q \\ a &\neq 0 \end{aligned}$$

$$\therefore 2y - 2ap^2 = (x - 2ap)(p+q)$$

$$2y - 2ap^2 = px + qx - 2ap^2 - 2apq$$

$$y = \frac{p+q}{2}x - apq$$

b) Since passes through focus, $(0, a)$ satisfies the equation

$$\text{i.e. } a = \frac{p+q}{2}(0) - apq$$

$$pq = -1$$

c) By midpoint formula

$$(1) \quad M \equiv \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right)$$

$$\equiv \left(a(p+q), \frac{a(p^2+q^2)}{2} \right)$$

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(ii) $N \equiv (a(p+q), -a)$

(iii) By midpoint formula

$$\begin{aligned} T &\equiv \left(a(p+q), \frac{\frac{a(p^2+q^2)}{2} - a}{2} \right) \\ &\equiv \left(a(p+q), \frac{a(p^2+q^2) - 2a}{4} \right) \\ &\equiv \left(a(p+q), \frac{a(p^2+q^2-2)}{4} \right) \end{aligned}$$

(iv) $x = a(p+q)$

$$\Rightarrow p+q = \frac{x}{a}$$

$$y = \frac{a(p+q)^2 - 2pq - 2}{4}$$

$$= \frac{a\left(\left(\frac{x}{a}\right)^2 - 2(-1) - 2\right)}{4}$$

$$= \frac{x^2}{4} \left(\frac{1}{a^2} \right) \quad a \neq 0$$

$$\Rightarrow x^2 = 4ay$$

Question 4

a) Let the terms be

$$a-d, a, a+d$$

Then

$$(a-d) + a + (a+d) = 3a$$

$$\text{and } 3a = 21 \quad (\text{given})$$

$$\therefore a = 7$$

$$(a-d)^2 + a^2 + (a+d)^2 = 155$$

$$a^2 - 2ad + d^2 + a^2 + a^2 + 2ad + d^2 = 155$$

$$3a^2 + 2d^2 = 155$$

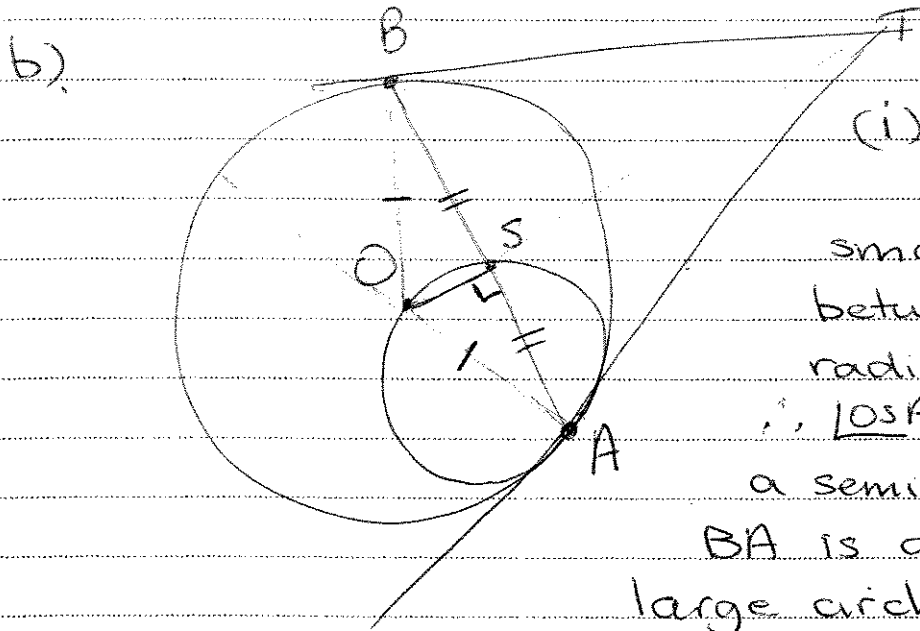
$$3(49) + 2d^2 = 155$$

$$2d^2 = 8$$

$$d^2 = 4$$

$$\therefore d = \pm 2$$

$\therefore$ Terms are 9, 7, 5 or 5, 7, 9.



(i) OA is a diameter of the small circle (angle between tangent and radius is 90°)

$\therefore \angle OSA = 90^\circ$ (angle in a semicircle)

BA is a chord of the large circle and OS is perpendicular to it $\Rightarrow$ OS bisects AB

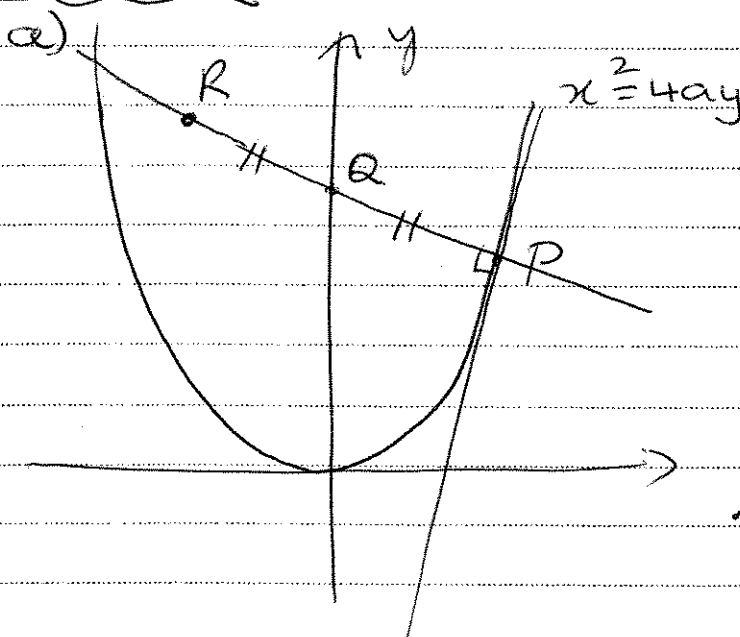
(ii) $\triangle BTA$ is isosceles (tangents from an external point equal)

$\angle TSA = 90^\circ$ (TS is an altitude of an isosceles triangle)

$$\therefore \angle OST = \angle OSA + \angle TSA = 180^\circ$$

$\therefore$ O, S, T are collinear.

Question 5



$$(i) x = 2at$$

$$\frac{dx}{dt} = 2a$$

$$y = at^2$$

$$\frac{dy}{dt} = 2at$$

$$\therefore \frac{dy}{dx} = \frac{2at}{2a} = t$$

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$$\therefore \text{gradient normal} = -\frac{1}{t}$$

Equation normal

$$y - at^2 = -\frac{1}{t}(x - 2at)$$

$$ty - at^3 = -x + 2at$$

$$x + ty - at^3 - 2at = 0$$

(ii) Put in $x=0$.

$$ty = at^3 + 2at$$

$$y = at^2 + 2a$$

$$t \neq 0.$$

$$\therefore Q \equiv (0, a(t^2 + 2))$$

(iii) Let $R \equiv (x_1, y_1)$

Then using midpoint formula.

$$0 = \frac{x_1 + 2at}{2} \Rightarrow x = -2at.$$

$$a(t^2 + 2) = \frac{y_1 + at^2}{2}$$

$$2a(t^2 + 2) = y_1 + at^2$$

$$\Rightarrow y_1 = at^2 + 4a$$

$$= a(t^2 + 4)$$

$$R \equiv (-2at, a(t^2 + 4))$$

(iv) From $x = -2at$

$$t = \frac{x}{-2a}$$

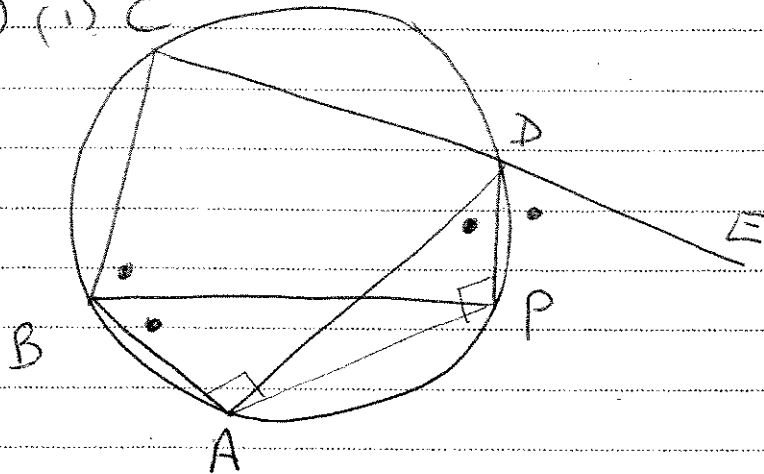
$$y = a\left(\frac{x^2}{4a^2} + 4\right)$$

$$y = \frac{x^2}{4a} + 4a$$

$$4ay = x^2 + 16a^2$$

$$x^2 = 4a(y - 4a)$$

b) (i) C



(ii) $\angle ABP = \angle ADP$ (angles standing on the same arc).

(iii) $\angle PDE = \angle CBD$ (exterior angle of a cyclic quadrilateral)

$\therefore \angle PDE = \angle ADP$ (since from (ii))

i.e. PD bisects $\angle ADE$.

(iv) Circle centre at intersection of BP and AD (as have two angles in a semicircle).

Question 6

(i) $A_1 = 30000(1 + 1.12) - m$

(ii) $A_2 = [30000(1.12) - m](1.12) - m$

$$= 30000(1.12)^2 - m(1.12 + 1)$$

(iii) $A_{20} = 30000(1.12)^{20} - m(1.12^{19} + 1.12^{18} + \dots + 1)$

$$= 30000(1.12)^{20} - m \frac{(1.12^{20} - 1)}{1.12 - 1}$$

(Using GP formula with $a=1$, $r=1.12$, $n=20$)

$A_{20} = 0$ since loan finished.

$$\therefore 30000(1.12)^{20} = \frac{m(1.12^{20} - 1)}{1.12 - 1}$$

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$$M = \frac{30000 (1.12)^{20} \times 12}{1.12^{20} - 1}$$

$$= \$ 4016.36.$$

$$b) \sum_{r=1}^n r^2 = \frac{1}{6} n(n+1)(2n+1).$$

$$\text{i.e. } 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6} n(n+1)(2n+1).$$

Let $n=1$

$$\text{LHS} = 1^2 = 1$$

$$\begin{aligned} \text{RHS} &= \frac{1}{6} (1)(1+1)(2 \times 1 + 1) \\ &= \frac{1}{6} (2)(3) \\ &= 1 = \text{LHS} \end{aligned}$$

 $\therefore$ True for $n=1$.Let $n=k$ and assume result true

$$\text{i.e. } \sum_{r=1}^k r^2 = \frac{1}{6} k(k+1)(2k+1).$$

Let $n=k$ and try to show result still holds

$$\text{i.e. } \sum_{r=1}^{k+1} r^2 = \frac{1}{6} (k+1)(k+2)(2k+3),$$

$$\text{LHS} = \sum_{r=1}^{k+1} r^2$$

$$= \sum_{r=1}^k r^2 + (k+1)^2$$

$$[S_{k+1} = S_k + T_{k+1}]$$

$$= \frac{1}{6} (k)(k+1)(2k+1) + (k+1)^2$$

using assumption.

$$\therefore \text{LHS} = (k+1) \left[\frac{1}{6} k(2k+1) + (k+1) \right]$$

$$= \frac{1}{6} (k+1) [2k^2 + k + 6(k+1)]$$

$$= \frac{1}{6} (k+1) [2k^2 + 7k + 6]$$

$$= \frac{1}{6} (k+1)(k+2)(2k+3)$$

 $\therefore$ If true for $n=k$ also true for $n=k+1$ Since true for $n=1$ also true for $n=2$ and so the induction hypothesis true for all